

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 09

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• The lowest ore purity that can be mined economically with existing technology • Deposit X (0.9% is above 0.4%; Y at 0.35% is part of the resource, not the reserves)	2
01.2	• X has higher grade so less rock mined, less spoil, less energy and pollution per tonne of copper • But X is deep: costs rise more than proportionally with depth (sides cannot be vertical so much more rock removed; more water inflow and pumping) • X has hard overburden requiring blasting (cost, noise, dust) and high rainfall increasing drainage costs • X is far from transport infrastructure so transport costs are high • Y is shallow with loose overburden (cheap to remove but landslide risk so gentler slopes / larger land take), near a railway, low rainfall • Y's grade is below the COOG so it is not viable now but would become viable if copper prices rose or extraction technology improved (eg bioleaching); both sulfides risk acid mine drainage	6
01.3	<i>Any 2 from:</i> • Higher market value / price of the metal • Improved extraction technology • Lower energy costs	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 138-139; AQA spec 3.2.3.5.	
02.1	• Organic (methyl) mercury	1
02.2	• Anaerobic microbes converted the inorganic mercury into organic methyl mercury • Methyl mercury is liposoluble so it is absorbed, crosses into the brain and biomagnifies along the food chain to fish eaten by people	2

Question	Marking guidance	Marks
02.3	• Most heavy metals are less soluble at high pH • So they remain immobile and are not leached into water or absorbed by organisms	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	
03.1	<i>Any 4 from:</i> • Topography: a narrow exit from a large deep basin so a small dam holds a large volume • Geology: impermeable rock beneath (no seepage), strong enough to support the dam, no faults / seismic activity • Catchment area: large enough with regular, reliable rainfall / river inflow; climate not too hot (evaporation losses) • Catchment land use: low pollution risk (industry, pesticides); low soil erosion (sedimentation reduces capacity) • Infrastructure / proximity to demand: access for construction, cost of transporting water • Existing land use: value of what is lost (urban areas, conservation sites, farmland) - loss:benefit analysis	4
03.2	• Advantage: fewer land-use conflicts as no land is flooded; large volume near coastal cities • Disadvantages: destroys / changes important intertidal habitats (mudflats for wading birds); obstacle to shipping; collects pollution from the whole river catchment; sedimentation	3
03.3	• Very energy-intensive and expensive (reverse osmosis / distillation) • Only worthwhile where seawater is available and freshwater supplies are inadequate (eg Malta, Middle East)	2
03.4	• Surplus water in wet periods (when infiltration capacity is exceeded and runoff would go to the sea) is pumped underground or diverted into lagoons to infiltrate, so abstraction in dry periods is balanced by recharge	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-124; AQA spec 3.2.2.5.	
04.1	• $10^3 / 6^3 = 1000 / 216$ • = 4.6 times (accept 4.63)	2
04.2	• C has the highest velocity (most energy, and energy rises with the cube of velocity), strong reliable winds over water, few land-use conflicts, but long expensive cable and offshore installation / maintenance costs • A is windy but on a migration route (bird strikes) and peat bog (foundations / tracks damage hydrology; carbon release); far from grid; scenic objections • B is close to the grid with good wind but near a village (noise, visual objection, shadow flicker) - possible with careful siting • D is cheap to connect and land can still be farmed between turbines, but low velocity gives little energy • Judgement, eg C or B; turbines spaced 3-5 blade diameters apart to avoid wind shadow	5

Question	Marking guidance	Marks
04.3	<i>Any 2 from:</i> • Driven by wind from any direction so no motor to turn them; suitable for turbulent urban sites • Generator can be at ground level so a lighter tower • Quieter • Operate at lower wind velocities • Blades more visible so fewer bird deaths	2
04.4	• No gearbox: fewer breakdowns, less friction so start generating at lower wind speeds, quieter, lighter nacelle (DC converted to AC by an inverter)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 206-209; AQA spec 3.3.4.1.	
05.1	• Image databases: identifying individuals (tiger stripes, whale shark spots, dolphin fins) to find territory size, movements, lifespan, social groups - few species have unique recognisable features • eDNA: detecting the presence of a species from DNA shed into water (great crested newt, invasive carp) - shows presence not abundance; contamination / DNA transport by water • Geolocator tags: tracking small birds on migration where transmitters are too heavy - only record light and time so position is approximate; animal must be recaptured to download data	6
05.2	• Microwaves emitted by the satellite reflect off the surface giving data on wave height, wind velocity, sea surface altitude and oil slicks (which smooth the surface) • Microwaves pass through cloud so monitoring is not interrupted by weather	2
05.3	<i>Any 2 from:</i> • Droppings: diet, gender, territory size • Owl pellets: diet • Feeding marks (chewed hazel nuts - dormouse), tracks / footprints, nests / burrows, scratching posts (territory)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-407; AQA spec 3.7.2.3.	
06.1	• $13 / 480 = 0.0271$ years • 0.0271×365 • = 10 days (accept 9.9-10)	3
06.2	• $1\,340\,000 / 425 = 3153$ years (accept 3100-3200)	1
06.3	• It has a very short residence time / very rapid transfer rate ($480 \times 10^{15} \text{ dm}^3$ per year) • So it moves large volumes of water each year, delivering precipitation to land; small reservoirs with fast transfers respond quickly to change	2
06.4	• The ocean is so large that an added or removed volume is a tiny proportion of the total • The same volume is a large proportion of a small reservoir so its level / residence time changes noticeably	2
06.5	• Water is constantly moving between reservoirs (evaporation, precipitation, runoff, infiltration, transpiration) • But inflows and outflows for each reservoir balance so the volume in each reservoir stays constant over long periods	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 115-116; AQA spec 3.2.2.1.	
07.1	• $25 + 5 = 30\%$	1
07.2	• UV: short wavelength; mostly absorbed in the stratosphere by monatomic, diatomic and triatomic oxygen (ozone layer) which are formed and destroyed in a dynamic equilibrium; converted to chemical energy in these reactions; little reaches the surface • Visible light: transmitted through the atmosphere easily (some reflected by clouds / dust); absorbed by the Earth's surface and converted to heat; some absorbed by chlorophyll and converted to chemical energy in photosynthesis • IR: emitted by the warm surface (long-wave); absorbed by greenhouse gases (CO₂, water vapour, methane) and converted to heat; the warm atmosphere re-emits IR to the surface and reduces conduction losses; some escapes to space	6
07.3	• Greenhouse gases absorb most of the IR emitted by the surface and re-radiate it in all directions, including back down • The surface is repeatedly warmed and re-emits, so the total IR exchanged between surface and atmosphere exceeds the incoming solar energy; only the net loss (70%) balances the absorbed solar energy at equilibrium	2
07.4	• The atmosphere forms layers (troposphere, stratosphere, mesosphere) with different temperature gradients because energy is absorbed at different altitudes (eg UV heating the stratosphere, surface heating the troposphere)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 87-89; AQA spec 3.2.1.1.	
08.1	• Minimum ozone fell steeply from about 225 DU in 1979 to about 95 DU in the mid 1990s • Since then it has fluctuated around 100-115 DU with a slight recovery / no further decline	2
08.2	• $(225 - 95) / 225 \times 100$ • = 57.8% (accept 58%)	2
08.3	• Stratospheric temperatures over Antarctica are the lowest on Earth so stratospheric clouds / ice crystals form • Ice crystal surfaces allow reactions that convert the chlorine released from CFCs into chlorine molecules (Cl₂) during the dark winter • The polar vortex (rotating winds) prevents mixing with the rest of the atmosphere, keeping temperatures low and chlorine concentrated • When sunlight returns in spring (September) it splits the chlorine molecules into free radicals that destroy ozone; levels recover by December as the vortex breaks down	4
08.4	• Ground and satellite UV / ozone measurements show depletion but not the chemistry / correlation is not causation • Air samples collected from the stratosphere by helium balloons and high-flying research aircraft detected chlorine and chlorine monoxide, confirming the Rowland-Molina hypothesis	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 110-111; AQA spec 3.2.1.3.	

Question	Marking guidance	Marks
09.1	- The human ear is sensitive to a very wide range of sound pressures / powers (about 10^{12} range) - A logarithmic scale compresses this into a convenient range of numbers: each 10 dB increase represents a ten-fold increase in sound intensity	2
09.2	- Difference = 30 dB = 3 x 10 dB = 10^3 = 1000 times	2
09.3	88 dB - Doubling the sound intensity increases the level by 3 dB (logarithmic scale)	2
09.4	0 dB is set as the quietest sound that can be heard (at 1000 Hz) - the threshold of human hearing; quieter sounds exist but cannot be heard	1
09.5	20 to 20 000 Hz - dB(A) weights sounds between about 1000 and 4000 Hz more heavily - Because the nerve cells / hair cells in the cochlea are more sensitive to these frequencies so they are heard more easily / cause more annoyance; range narrows with age or noise damage	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 286-287; AQA spec 3.4.3.2.13.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - Organic nutrients (sewage, manure, silage effluent, paper mill / tannery / food processing effluent) are food for aerobic microbes whose respiration deoxygenates the water, killing fish and invertebrates; bacterial growths cover the bed, blocking light - Pathogens in sewage (cholera, typhoid, dysentery) spread via contaminated water; decay of organic matter releases inorganic nutrients - Inorganic nutrients: nitrates (fertiliser / manure runoff, leaching) and phosphates (sewage effluent, detergents, eroded soil) cause cultural eutrophication: algal blooms, shading of macrophytes, cyanobacterial toxins, night-time and end-of-summer deoxygenation - Human health: nitrites and methaemoglobinaemia; possible carcinogen via nitrosamines - Control of organic wastes: sewage treatment (pre-treatment, primary sedimentation removing >95% organic matter, aeration / activated sludge or trickling filters, anaerobic digestion of sludge with biogas) - Control of nutrients: tertiary phosphate stripping with iron(III) sulfate, denitrification in filter beds, micro-strainers / UV; Nitrate Vulnerable Zones, fertiliser timing / amount, low-solubility fertilisers, legumes, manure management; dredging sediments - Evaluation: sewage treatment has virtually eliminated deoxygenation and waterborne disease in developed countries but storm overflows and combined sewers still discharge untreated sewage; tertiary treatment is expensive so only used at sensitive sites; diffuse agricultural nitrate is hard to control and aquifers respond slowly; phosphate legacy in sediments - Judgement on point-source (sewage) control being more successful than diffuse-source (agriculture) control	9
10.2	Algae / cyanobacteria absorb nutrients directly from the water (not the sediments) so they use dissolved nutrients more easily and grow rapidly	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 266-272; AQA spec 3.4.3.2.9.	

Question	Marking guidance	Marks
11.1	Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Attributes: accuracy, precision, representative samples, reliability, anomalies, validity; hypothesis and null hypothesis; preliminary studies • Sampling design: random vs systematic location and avoiding bias; timing (diurnal, seasonal, long-term, weather); sample size and number and the running mean; standardisation of technique and control of other variables • Techniques for abiotic and biotic factors and their limitations (quadrats, transects, traps, mark-release-recapture assumptions, keys, meters and calibration) • Statistics: variability, standard deviation and error bars, significance and p-values, correlation vs causation; choosing Spearman's rank, chi-squared, t-test, Mann-Whitney U • Application to Paper 1 topics: monitoring pollution (noise, air quality, water), climate proxies, energy surveys, soil analysis • Evaluation of trade-offs (cost, time, disturbance) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.	25

Question	Marking guidance	Marks
11.2	Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Historical data limitations; proxy data - ice cores, isotope ratios (O-18:O-16, C-14), Mg:Ca in calcite, tree rings, pollen, coral • Satellite sensors: visible light imagery, infrared for vegetation and temperature, radar microwaves for wind, wave height, sea level and oil, weather monitoring; passing through cloud • Sensor carriers: ROVs / drones, AUVs, balloons, aircraft, animals; data recorders for abiotic factors; GPS tracking of floats and sampling sites; Argo floats and ocean monitoring • Climate models and their assumptions; monitoring atmospheric composition (Mauna Loa), ozone (Dobson units), sea level and ice • Applications to hydrosphere, minerals (remote sensing surveys), soils, pollution monitoring • Evaluation: improved coverage and resolution vs cost, calibration, interpretation, remaining uncertainty; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 9, 89-107, 402-407; AQA spec 3.2, 3.7.	25